

Explainable Fake Review Detection for Bangla and Banglish E-Commerce Text

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Abstract—Fake-review detection in Bangla and Banglish e-commerce text is difficult because labeled data are limited, class distributions are imbalanced, and reviews combine Bangla, English, transliteration, numerals, emojis, and promotional language. This paper presents an explainable, CPU-feasible PySpark pipeline that fuses TF-IDF, Word2Vec, stylometric, promotional, script, and duplicate-structure signals. The final run uses duplicate-aware splitting, class-weighted supervised learning, and confidence- and class-capped pseudo-labeling on an unlabeled BanglishRev pool. On a locked test set of 1,356 BFRD reviews, the selected Random Forest achieves macro-F1 0.9194, fake-review PR-AUC 0.9213, and accuracy 0.9617 using gold labels only. Adding 31,605 capped pseudo-labels decreases macro-F1 to 0.9098 and fake-review PR-AUC to 0.9033. The result supports multi-signal explainable detection while showing that confidence filtering does not remove confirmation bias or domain-shift risk.

Index Terms—Bangla NLP, fake review detection, opinion spam, code-mixed text, pseudo-labeling, explainable machine learning, PySpark

I. INTRODUCTION

Online reviews influence purchase decisions, seller reputation, and platform trust. Artificially generated or coordinated reviews therefore create both economic and social harm. The problem is especially challenging for Bangla and Banglish e-commerce text: the same review can contain Bengali Unicode, Latin transliteration, English product terms, prices, emojis, URLs, informal spelling, and promotional slogans. The project goal is a binary detector that is accurate enough for triage, auditable by analysts, and feasible on CPU-oriented infrastructure.

The completed system treats fake-review detection as an evidence-fusion problem rather than a single text-classification task. It combines lexical evidence, dense sentence-level representations, stylometric and promotional signals, and duplicate structure. It also tests whether an unlabeled BanglishRev pool can improve the classifier through controlled pseudo-labeling. The contribution is therefore an integrated and auditable methodology, not a new classifier. The main contributions are:

- a Bangla/Banglish multi-signal feature pipeline with explicit privacy-safe text views;
- duplicate-aware splitting and exact/near-duplicate audits to reduce leakage;
- confidence-, class-, and volume-capped pseudo-labeling with reduced sample weight;
- global and local explanations through tree importance, linear coefficients, error analysis, and LIME; and

TABLE I
FINAL-RUN DATA INVENTORY AND SPLIT PROTOCOL.

Partition	Fake (0)	Genuine (1)	Total
Gold train	934	5,387	6,321
Validation	201	1,155	1,356
Locked test	201	1,155	1,356
Unlabeled feature pool	–	–	322,954

- a locked-test comparison showing that a validation gain from semi-supervision did not transfer to the final test set.

II. DATA AND PROBLEM FORMULATION

Let each cleaned review be x_i with binary label $y_i \in \{0, 1\}$, where $y_i = 0$ denotes fake and $y_i = 1$ denotes genuine. The gold source is the Bengali Fake Review Dataset (BFRD); the auxiliary unlabeled source is BanglishRev. The final run scanned 1,746,943 BanglishRev written reviews and retained 322,954 clean rows for feature processing. BFRD contains 9,049 raw rows and 9,034 clean rows after validation and duplicate checks.

The locked test is evaluated only after model selection. Exact duplicate hashes are audited, and MinHash locality-sensitive hashing (LSH) is used for near-duplicate analysis. Supervised transformations remain train-scoped; the final feature audit records a transductive global fit for unsupervised vocabulary/embedding construction, which is reported explicitly as a validity caveat.

III. PIPELINE AND METHODOLOGY

Fig. 1 gives the complete pipeline. The design separates acquisition, cleaning, leakage control, representation learning, supervised selection, semi-supervised expansion, and explanation.

A. Preprocessing and Privacy-Safe Text Views

The cleaning stage validates schemas, removes invalid rows, normalizes Unicode and whitespace, and records provenance. Separate text views support privacy-aware analysis: email, phone, URL, price, and mention patterns can be removed from model text, while source identifiers and images are excluded. Tokenization is deterministic and shared by the final evaluation and LIME modules. The feature audit records the following handcrafted variables: character and token length; code-mix

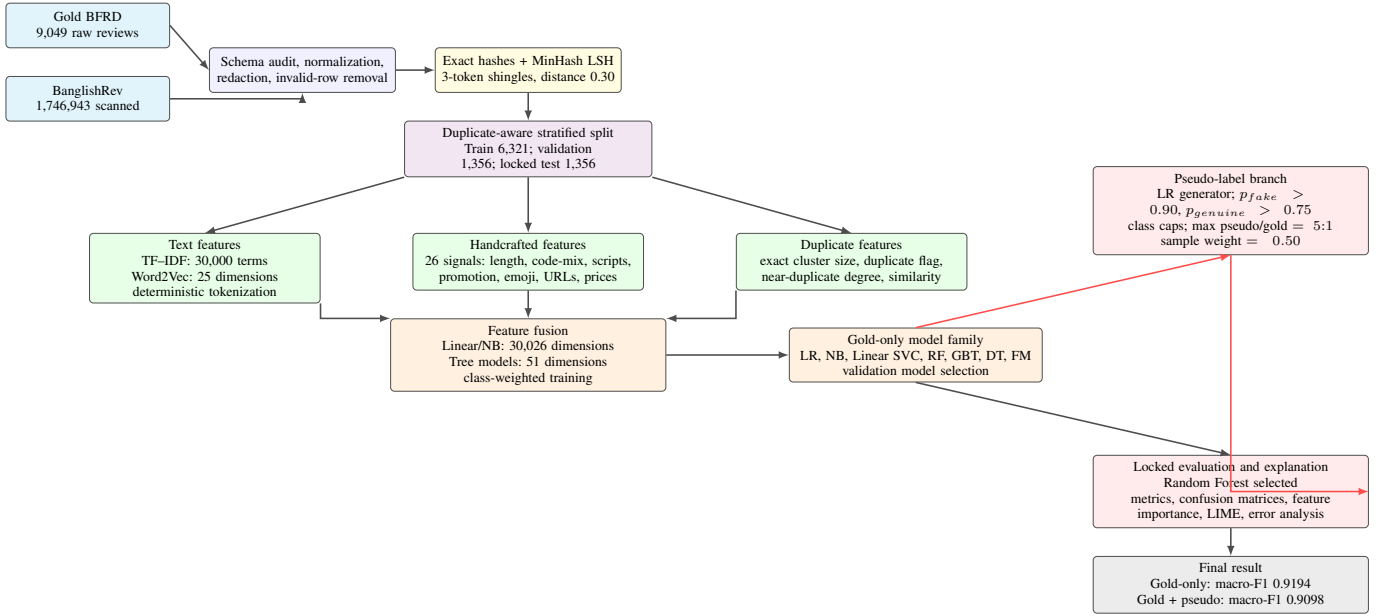


Fig. 1. Detailed final-run pipeline and methodology. Red edges denote the controlled pseudo-label branch; the test split remains locked and is never pseudo-labeled.

TABLE II
FEATURE GROUPS AND MODEL MAPPING.

Group	Configuration	Main use
TF-IDF/count	30,000 vocabulary terms	LR, Linear SVC, NB
Word2Vec	25-dimensional mean vector	RF, GBT, DT, FM
Stylometry	lengths, scripts, code-mix, repeated chars	all tree/linear fusion
Behavioral	promotion, emoji, URLs, prices, mentions	all tree/linear fusion
Duplicate graph	exact and MinHash-derived statistics	audit and ablation

ratio; repeated-character ratio; Bengali, Latin, digit, and other-character counts; URL, email, phone, price, hashtag, mention, exclamation, and emoji counts; promotional-keyword hits; exact-duplicate cluster size and flag; near-duplicate degree, similarity, and cluster size; and binary presence flags.

B. Lexical and Dense Features

For a term t and document d , term frequency-inverse document frequency is computed as

$$\text{tfidf}(t, d) = \text{tf}(t, d) \log \left(\frac{N + 1}{\text{df}(t) + 1} \right), \quad (1)$$

where N is the number of documents and $\text{df}(t)$ is the document frequency. The final Kaggle-safe run uses a 30,000-term count vocabulary; TF-IDF and non-negative count features are used for linear and Naive Bayes models.

Word2Vec produces a 25-dimensional review representation by averaging token vectors. For review tokens $w_1, \dots, w_m$

with embeddings $v(w_j)$,

$$z_d = \frac{1}{m} \sum_{j=1}^m v(w_j), \quad z_d \in \mathbb{R}^{25}. \quad (2)$$

The model-ready dimensions are 30,026 for linear/Naive Bayes features and 51 for tree features, including 26 handcrafted variables.

C. Duplicate Structure

For token-shingle sets A and B , the MinHash similarity proxy is based on Jaccard similarity,

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}, \quad (3)$$

with a distance threshold of 0.30, three hash tables, 8,192 hash dimensions, and three-token shingles. For each review, graph features summarize local duplicate exposure without directly copying review text.

D. Class-Weighted Models

The model family includes Logistic Regression (LR), Multinomial Naive Bayes (NB), Linear SVC, Random Forest (RF), Gradient-Boosted Trees (GBT), Decision Tree (DT), and Factorization Machine (FM). For a linear classifier, the probability of a fake review is

$$p_{\text{fake}}(x) = p(y = 0 | x) = \sigma(w^\top x + b) = \frac{1}{1 + e^{-(w^\top x + b)}}. \quad (4)$$

The weighted LR objective is

$$\min_{w, b} \frac{\lambda}{2} \|w\|_2^2 + \sum_i \alpha_{y_i} \log \left(1 + e^{-y_i (w^\top x_i + b)} \right), \quad (5)$$

where α_{y_i} is the class weight. In the final run, weights are $\alpha_0 = 3.38383$ for fake reviews and $\alpha_1 = 0.58669$ for genuine reviews. RF combines trees using

$$\hat{p}_{\text{fake}}(x) = \hat{p}(y = 0 \mid x) = \frac{1}{T} \sum_{t=1}^T h_t(x), \quad (6)$$

where h_t is the probability estimate of tree t .

E. Controlled Pseudo-Labeling

The tuned LR model generates probabilities on the unlabeled pool. A pseudo-label is accepted only when

$$\tilde{y}(x) = \begin{cases} 0, & p_0(x) > 0.90 \\ 1, & p_1(x) > 0.75 \\ \emptyset, & \text{otherwise,} \end{cases} \quad (7)$$

then class caps and a maximum pseudo/gold ratio of 5:1 are applied. Accepted pseudo-labels receive sample weight $\omega_p = 0.50$, while gold examples retain weight 1. The primary run accepted 304,667 candidates before capping and 31,605 after capping: 6,321 fake and 25,284 genuine. A sensitivity branch uses a fake threshold of 0.95 and genuine threshold of 0.70.

F. Explainability

Global evidence is obtained from LR coefficients and tree feature importance. Local evidence is generated with LIME, which fits a local surrogate g around a point x by minimizing

$$\xi(x) = \arg \min_{g \in \mathcal{G}} L(f, g, \pi_x) + \Omega(g), \quad (8)$$

where f is the RF predictor, π_x weights perturbed samples by proximity to x , and Ω controls surrogate complexity. The final audit contains 110 LIME cases.

IV. EXPERIMENTAL DESIGN AND METRICS

The gold split contains 6,321 training, 1,356 validation, and 1,356 locked-test rows. Model selection uses the gold validation split; the final test is scored once for gold-only and gold-plus-pseudo configurations. Accuracy is reported but not used alone because the fake class is imbalanced. For class c , precision, recall, and F1 are

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}, \quad F1_c = \frac{2P_c R_c}{P_c + R_c}. \quad (9)$$

Macro-F1 averages $F1_c$ across both classes. Fake PR-AUC summarizes precision–recall behavior for the fake class; ROC-AUC is included as a secondary ranking metric.

V. RESULTS

A. Validation Comparison

Table III reports the gold validation comparison. RF is the best gold-only model by macro-F1 and fake PR-AUC in the final run. Expanded training improves Linear SVC and NB but decreases RF, LR, and GBT, illustrating that pseudo-labeling effects are model-dependent.

TABLE III
GOLD AND EXPANDED VALIDATION PERFORMANCE.

Model	Gold F1	Exp. F1	$\Delta F1$	Exp. PR-AUC
RF	0.9252	0.9224	-0.0028	0.9025
LR	0.9094	0.9078	-0.0016	0.8843
GBT	0.9074	0.9042	-0.0033	0.8984
Linear SVC	0.7758	0.8786	+0.1028	0.8590
DT	0.8650	0.8690	+0.0040	0.8630
NB	0.7139	0.7501	+0.0362	0.5592
FM	0.6673	0.6597	-0.0077	0.3609

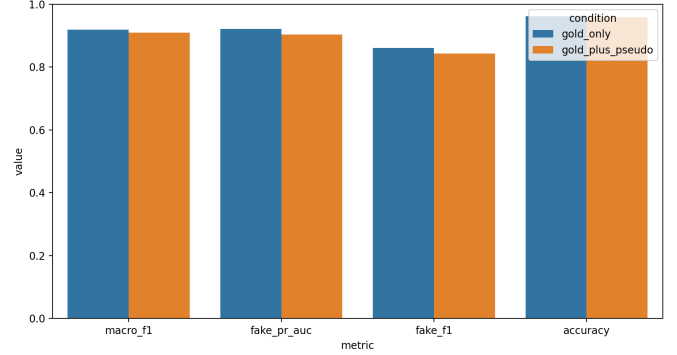


Fig. 2. Final-run comparison of model and training conditions.

B. Locked-Test Evaluation

Table IV uses the machine-readable final evaluation artifact. Gold-only RF is the selected final condition. Its confusion matrix is $\begin{bmatrix} 161, & 40 \\ 12, & 1143 \end{bmatrix}$, where rows are true labels and columns are predictions. Gold-plus-pseudo RF changes this to $\begin{bmatrix} 154, & 47 \\ 10, & 1145 \end{bmatrix}$.

The pseudo-label configuration reduces macro-F1 by 0.0096 and fake PR-AUC by 0.0181. It also reduces fake recall from 0.8010 to 0.7662, while genuine recall increases from 0.9896 to 0.9913. Thus, the capped pseudo-labels make the model slightly more conservative toward the genuine class but less sensitive to fake reviews.

C. Ablation and Error Analysis

Table V shows that TF-IDF alone is substantially weaker than the full feature configuration. Removing near-duplicate features changes LR macro-F1 by only +0.0013 in gold validation and leaves expanded performance unchanged, so the current experiment supports their value as an audit signal more strongly than as a dominant predictive component.

The final error analysis contains 1,299 correct predictions, 47 false negatives, and 10 false positives. False negatives are shorter on average (450.4 characters and 69.0 tokens) and have higher promotional-hit rates (0.723) than correct cases (674.7 characters, 101.5 tokens, 0.379 promotional hits). False positives are longer (746.3 characters) and always contain at least one promotional hit in the recorded aggregate. This suggests that concise promotional fakes and legitimate promotional language are both difficult cases.

TABLE IV
LOCKED-TEST RESULTS FOR THE SELECTED RANDOM FOREST.

Condition	Macro-F1	Acc.	Fake PR-AUC	Fake F1	Fake Recall
Gold only	0.9194	0.9617	0.9213	0.8610	0.8010
Gold+pseudo	0.9098	0.9580	0.9033	0.8438	0.7662
Δ	-0.0096	-0.0037	-0.0181	-0.0171	-0.0348

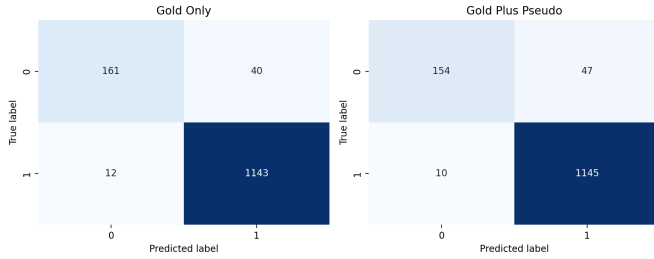


Fig. 3. Confusion matrices for the two locked-test conditions.

D. Decision-Support Analytics

The final PySpark SQL notebook was executed against the saved feature and evaluation artifacts and wrote six operational outputs under `decision_support_outputs`. These outputs translate model results into defensible analytics decisions rather than autonomous enforcement decisions.

The review queue is for analyst prioritization. It is not proof that a review or seller is fraudulent. The confusion matrices show that pseudo-label expansion misses seven additional fake reviews while reducing two false alarms, so the lower fake-review recall is operationally more important than the small reduction in false positives. The recommended action is human review with explanations, policy evidence, and an appeal path. Drift and duplicate statistics are monitoring signals, not causal evidence.

VI. EXPLORATORY DATA ANALYSIS AND INTERPRETATION

The project artifacts include class-distribution, length/code-mix, handcrafted-signal, and correlation visualizations. The class distribution motivates macro metrics and class weighting. The length and code-mix plots show why character-level and script-aware signals complement lexical features. The correlation heatmap motivates retaining both raw counts and binary presence features only with care, because correlated signals can inflate tree importance.

Global explanations rank emoji presence/count, latent Word2Vec dimensions, promotional presence, Bengali-character count, and promotional-keyword hits among influential Random Forest inputs. These rankings are associations, not causal rules. LIME explanations provide local evidence for 110 test cases, but their words should be reviewed for privacy before publication and interpreted as surrogate explanations rather than model internals.

VII. THREATS TO VALIDITY, ETHICS, AND LIMITATIONS

The detector inherits BFRD annotation and collection bias and has not been externally validated on a second la-

TABLE V
FEATURE ABLATION ON VALIDATION DATA.

Cond.	Features	Macro-F1	PR-AUC	Fake F1
Gold	Full	0.9094	0.8899	0.8446
Gold	TF-IDF only	0.7383	0.5919	0.5594
Gold	No near-dup.	0.9107	0.8901	0.8468
Expanded	Full	0.9078	0.8843	0.8413
Expanded	TF-IDF only	0.7240	0.5938	0.5551
Expanded	No near-dup.	0.9078	0.8843	0.8413

TABLE VI
FINAL DECISION-SUPPORT EVIDENCE AND OPERATIONAL USE.

Output	Final evidence	Decision use
Class balance	6,321 train; 1,356 validation; 1,356 test. Test has 201 fake and 1,155 genuine.	Use class-aware metrics and weights.
Test comparison	Gold-only: macro-F1 0.9194, fake PR-AUC 0.9213, fake recall 0.8010. Expanded: 0.9098, 0.9033, 0.7662.	Keep gold-only as the default condition.
Review queue	All 1,356 test predictions ranked by normalized fake-risk score from Random Forest raw class votes.	Prioritize human investigation.
Error profile	1,299 correct, 47 false negatives, and 10 false positives. False negatives average 450.4 characters and 0.723 promotional hits.	Collect more short promotional examples.
Duplicate exposure	Four exact-duplicate test rows; mean exact cluster size 1.0029.	Retain duplicate auditing and leakage controls.
Drift indicators	Train/validation/test means remain close for length, tokens, code-mix, promotion, emoji, and promo flags.	Continue monitoring; no large split-level shift is indicated.

beled source. BanglishRev is unlabeled and may differ from BFRD, so pseudo-labeling can amplify domain mismatch. The recorded feature audit also reports a global unsupervised fit across gold and unlabeled rows; future releases should make the transductive scope explicit and add a strictly train-fitted comparison. The final run reports point estimates only; bootstrap intervals or repeated splits would strengthen the conclusions.

Deployment should be human-in-the-loop. A prediction should trigger review or ranking, not irreversible seller sanctions, because false positives can damage legitimate businesses and false negatives can leave deceptive content online. Review text may contain personal information; stored artifacts should use hashes or redacted examples, and LIME displays should be screened before release.

VIII. CONCLUSION

This report documents a reproducible, explainable Bangla/Banglish fake-review detection pipeline built from the final outputs and final run. The selected gold-only Random Forest reaches 0.9194 macro-F1 and 0.9213 fake PR-AUC on the locked test split. The multi-signal feature design is strongly supported by the TF-IDF ablation, while near-duplicate fields are currently more defensible as leakage-control and audit features than as independently validated performance drivers. Most importantly, controlled pseudo-labeling decreases locked-test performance in this run despite useful gains for some validation model families. Future work should add external evaluation, uncertainty intervals, stronger Bangla tokenization, calibration, and human-reviewed pseudo-label audits.

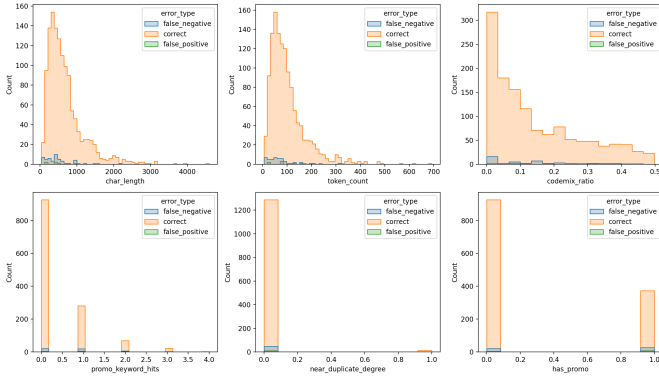


Fig. 4. Final-run error-analysis visualization.

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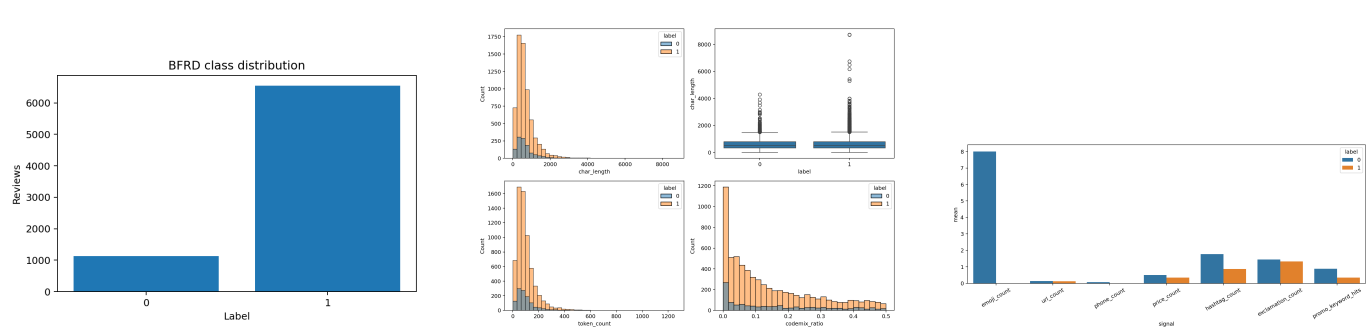


Fig. 5. Selected exploratory visualizations from the final output: class distribution, length/code-mix behavior, and handcrafted signals.

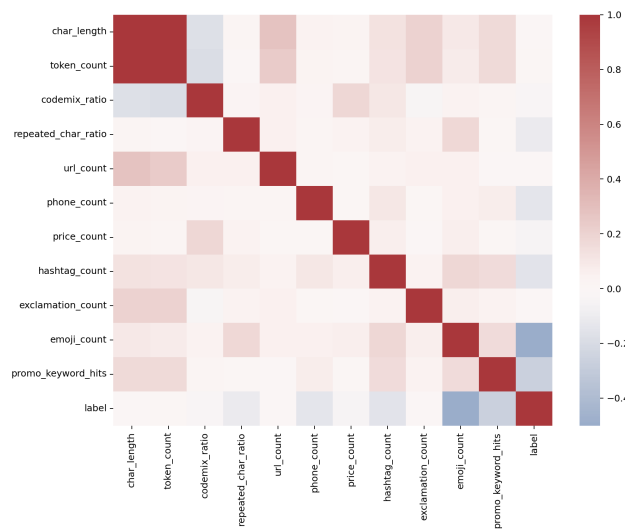


Fig. 6. Correlation structure among selected handcrafted signals.